

■ Python Interview Cheatsheet

■ Linked List - Reverse a Linked List

```
Code:
class ListNode:
    def __init__(self, val=0, next=None):
        self.val = val
        self.next = next

def reverseList(head):
    prev, curr = None, head
    while curr:
        nxt = curr.next
        curr.next = prev
        prev, curr = curr, nxt
    return prev
```

Example: Input = 1 -> 2 -> 3 -> 4 -> None
Dry Run:
Step 1: 1->None
Step 2: 2->1->None
Step 3: 3->2->1->None
Step 4: 4->3->2->1->None
Output: 4->3->2->1->None

■■ Stack - Largest Rectangle in Histogram

```
Code:
def largestRectangleArea(heights):
    stack, max_area = [], 0
    for i, h in enumerate(heights + [0]):
        while stack and heights[stack[-1]] > h:
            height = heights[stack.pop()]
            width = i if not stack else i - stack[-1] - 1
            max_area = max(max_area, height * width)
        stack.append(i)
    return max_area
```

Example: heights = [2,1,5,6,2,3]
Dry Run:
- At 5,6: stack grows
- At 2: pop 6, pop 5 -> max area = 10
Final Output: 10

■ Queue - Sliding Window Maximum

```
Code:
from collections import deque
def maxSlidingWindow(nums, k):
    dq, res = deque(), []
    for i, num in enumerate(nums):
        while dq and dq[0] <= i - k:
            dq.popleft()
        while dq and nums[dq[-1]] < num:
            dq.pop()
        dq.append(i)
        if i >= k - 1:
            res.append(nums[dq[0]])
    return res
```

Example: nums = [1,3,-1,-3,5,3,6,7], k=3

Dry Run -> Output: [3,3,5,5,6,7]

■■ Heap - Kth Largest Element

```
Code:
import heapq
def findKthLargest(nums, k):
    heap = []
    for num in nums:
        heapq.heappush(heap, num)
        if len(heap) > k:
            heapq.heappop(heap)
    return heap[0]
```

Example: nums = [3,2,1,5,6,4], k=2
Dry Run -> Final Heap = [5,6]
Output = 5